

HINTS & SOLUTION

Q.1 (B)

$$K_b = \frac{MRT_b^2}{1000 \Delta H_{\text{vap}}}$$

$$K_b \propto M$$

As molecular mass increases K_b increases,

$$\therefore K_{b\text{CCl}_4} > K_{b\text{C}_6\text{H}_6} > K_{b\text{CS}_2} > K_{b\text{C}_2\text{H}_5\text{--O--C}_2\text{H}_5}$$

Q.2 (A)

Q.3 (B)

Q.4 (A)

Q.5 (A)

Q.6 (C)

Q.7 (D)

Q.8 (A)

Q.9 (B)

Q.10 (A)

Q.11 (C)

Q.12 (C)

Q.13 (C)

Q.14 (D)

Q.15 (A)

Q.16 (C)

Q.17 (C)

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Q.18 (A)

A = 10, B = 20, C = 5, $P_A^\circ = 100$ torr, $P_B^\circ = 90$ torr

$$\begin{array}{ccc} 6A & \rightarrow & A_6(s) \\ 10 & & 0 \end{array}$$

$$\begin{array}{ccc} 10-x & & \frac{x}{6} \end{array}$$

$$x = Kt = 10^{-1} \times 60 = 6$$

$$\text{after 1 hrs} \Rightarrow A = 4, B = 20, \text{Solute} = C + \frac{x}{6} = 5 + 1 = 6$$

$$P^\circ = P_A^\circ \times A \times P_B^\circ \times B$$

$$P^\circ = 100 \times \frac{4}{24} + \frac{20}{24} \times 90 = 16.66 + 75$$

$$P^\circ = 91.66 \text{ torr}$$

$$\frac{P^\circ - P^s}{P^s} = \frac{n}{N} \Rightarrow \frac{91.66 - P^s}{P^s} = \frac{6}{24} = \frac{1}{4}$$

$$P^s = 73.33 \text{ torr} = \frac{220}{3}$$

Q.19 (C)

After 100 min. A will be zero

$$\begin{array}{ccc} 6A & \rightarrow & A_0 \\ 10 & & 0 \\ 0 & & \frac{10}{6} \end{array}$$

$$A = 0, B = 20 \text{ Solute} = 5 + \frac{10}{6} = 6.66$$

After 100 min.

$$\frac{P^\circ - P^s}{P^s} = \frac{n}{N} \Rightarrow \frac{90 - P^s}{P^s} = \frac{6.66}{20}$$

$$P^s = 67.5 \text{ torr}$$

Q.20 (D)

Q.21 (B)

Q.22 (B)

Q.23 (A) (C)

(A) On increasing temperature K_H will increase

(B) Solubility of O_2 will be higher than H_2 at same temperature

(C) If sealed under low pressure, less CO_2 will evolve due to low solubility.

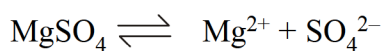
(D) Under water due to high pressure solubility of gases will also be high.

Q.24 4
(1), (2), (9), (10) Follow Raoult's law

Q.25 0377

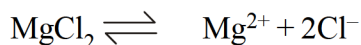
$$\Delta T_{b, \text{actual}} = i \times \Delta T_{b, \text{theoretical}}$$

For 'i' calculation



0.1

$$0.2 \quad 0.08 \quad 0.08 \Rightarrow n_{\text{Total}} = 0.18$$



0.1

$$0.04 \quad 0.06 \quad 0.12 \Rightarrow n_{\text{total}} = 0.22$$

$$i = \frac{0.18 + 0.22}{0.2} = 2$$

$$\Delta T_{b, \text{theo.}} = K_b \times m$$

$$= 0.785 \times \left[\frac{\frac{12}{120} + \frac{9.5}{95}}{78.5} \right] \times 1000 = 2$$

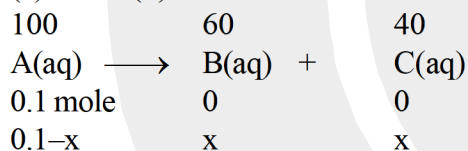
$$\Delta T_{b, \text{actual}} = 2 \times 2 = 4$$

$$T_b - T_b^\circ = 4$$

$$T_b = 104^\circ\text{C} = 377 \text{ K}$$

Q.26 $M = 53.8$

Q.27 (a) 66.66 (b) 50%



$$\Delta T_f = i K_f m$$

$$0.8533 = \frac{(0.1+x)}{0.1} \times 0.512 \times \frac{0.1}{90} \times 1000$$

$$x = 0.05$$

$$\% \text{ dissociation} = 50\%$$

$$M_{\text{avg}} = \frac{10}{0.15} = 66.66$$

Q.28 $a = 0.7333$

Q.29 $5.4 \times 10^5 \text{ g/mol}$

Q.30 $i = 2.5$

Q.31 $x = 25.6, y = 42.6$

Q.32 38.71 g



Q.33 1.0×10^{-4}

Q.34 0.73

Q.35 (a) 0.47, (b) 565 torr, (c) $x_A = 0.08$, $x_B = 0.92$, (d) 675 torr, (e) $x'_A = 0.11$, $x'_B = 0.89$

Q.36 92 mol% toluene; 96.8 mol % toluene

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